Naga Venkat

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Sr Data Engineer

PROFESSIONAL SUMMARY:

- ✓ Around 9 years of Full Software Development Life Cycle (SDLC) experience in Software, System Analysis, Design, Development, Testing, Deployment, Maintenance, Enhancements, Re-Engineering, Migration, Troubleshooting and Support of multi-tiered web applications in high performing environments.
- ✓ Hands-on experience in Hadoop Framework and its ecosystem including HDFS Architecture, Map Reduce Programming, Hive, Pig, Sqoop, HBase, Zookeeper, Couchbase, Storm, Solr, Oozie, Spark, Scala, Flume, Strom, and Kafka.
- ✓ Good experience in installing, configuring, and administrating Hadoop cluster of major Hadoop distributions Hortonworks, and Cloudera.
- ✓ Experience in batch processing and writing programs using Apache, Spark for handling real-time analytics and real Streaming of data.
- ✓ Good understanding of Zookeeper and Kafka for monitoring and managing Hadoop jobs and used ClouderaCDH4, CDH5 for monitoring and managing Hadoop cluster.
- ✓ Experience in Analysis, Design, Development and Big Data in Scala, Spark, Hadoop, Pig and HDFS environment.
- ✓ Good experience with design, coding, debug operations, reporting and data analysis utilizing python and using Python libraries to speed up development.
- ✓ Expertise in deploying Hadoop Architecture and various components such as HDFS, Job Tracker, Task Tracker, Name Node, Data Node, and Map Reduce/ Yarn concepts.
- ✓ Experience in loading the data into Spark RDD and performing in-memory data computation to generate the output responses.
- ✓ Good working experience on Spark (Spark Core Component, Spark SQL, Spark Streaming, Spark MLlib, Spark GraphX, SparkR) with Scala and Kafka.
- ✓ Understanding of Spark Architecture including Spark SQL, Data Frames, Spark Streaming, experience in analyzing Data with Spark while using Scala.
- ✓ Hands on experience in using other Amazon Web Services like S3, VPC, EC2, Autoscaling, RedShift, DynamoDB, Route53, RDS, Glacier, EMR.
- ✓ Experience on moving raw data between different systems using Apache NIFI.
- ✓ Experience in Datacenter Migration, Azure Data Factory (ADF) V2. Managing Database, Azure Data Platform services (Azure Data Lake (ADLS), Data Factory (ADF), Data Lake Analytics, Stream Analytics, Azure SQL DW, and HDInsight/Databricks.
- ✓ Experience in developing and designing POCs using Scala and deployed on the Yarn cluster, compared the performance of Spark with Hive and SQL/ Teradata.
- ✓ Strong understanding of RDD operations in Apache Spark i.e., Transformations, Actions, Persistence (Caching), Accumulators, Broadcast Variables, and Optimizing Broadcasts.
- ✓ In-depth understanding of Apache spark job execution Components like DAG, lineage graph, Dag Scheduler, Task scheduler, Stages, and task.
- ✓ Experience in migrating Python Machine learning modules to scalable, high-performance, and fault-tolerant distributed systems like Apache Spark.
- ✓ Expertise in developing multiple Spark jobs in Scala for data cleaning, pre-processing and aggregating.
- ✓ Expertise in working with Hive data warehouse tool-creating tables, data distribution by implementing partitioning and bucketing, writing and optimizing the HiveQL queries.
- ✓ Excellent understanding and knowledge of job workflow scheduling and locking tools/ services like Oozie and Zookeeper.
- ✓ Knowledge of ETL methods for data extraction, transformation and loading in corporate-wide ETL Solutions and Data Warehouse tools for reporting and data analysis.
- ✓ Comprehensive knowledge in Debugging, Optimizing and Performance Tuning of DB2, Oracle and MYSQL databases.
- ✓ Good experience with Informatica, AWS Glue for designing ETL Jobs for Processing of data.
- ✓ Experience with Azure transformation projects and implement ETL and data movement solutions using Azure Data Factory (ADF), and SSIS.
- ✓ Good understanding and knowledge of NoSQL databases like MongoDB, HBase and Cassandra.
- ✓ Experience in Monitoring Hadoop clusters using tools like Nagios, CloudWatch, and Cloudera Manager.

- ✓ Experience with operating systems such as Windows, Linux, RedHat, and UNIX.
- ✓ Experienced and skilled Agile Developer with a strong record of excellent teamwork and successful coding.
- ✓ Strong Problem Solving and Analytical skills and abilities to make Balanced Independent decisions.
- ✓ Exceptional ability to quickly master new concepts and capable of working in-group as well as independently with excellent communication skills.

TECHNICAL SKILLS:

- ✓ Operating Systems Linux, UNIX, Windows.
- ✓ Development Methodologies Agile/ Scrum, Waterfall.
- ✓ IDEs: Eclipse, Net Beans, IntelliJ
- ✓ Big Data Platforms: Hortonworks, Cloudera CDH4, CDH5
- ✓ Programming Languages: Python, Scala, SQL, Python NumPy, SciPy, Pandas, NLTK, Matplotlib, Beautiful Soup, Text Blob
- ✓ Hadoop Components: HDFS, Sqoop, Hive, Pig, MapReduce, YARN, Impala, Hue, Zookeeper
- ✓ Spark Modules: Spark Core Component, Spark SQL, Spark Streaming, Spark MLlib, Spark GraphX, SparkR
- ✓ RDBMS/NOSQL Databases: Oracle, MYSQL, Microsoft SQL Server, HBase, MongoDB, Cassandra
- ✓ Cloud Technologies: Amazon Web Services (AWS)
- ✓ AWS Services: EC2, IAM, S3, Autoscaling, CloudWatch, Route53, EMR, RedShift, Glue MS Azure SQL DW, HDInsight, Databricks, Azure Data Factory
- ✓ DevOps Tools: Ant, Maven, Jenkins, Git, GitHub
- ✓ Visualization and Reporting: Power BI, Tableau

PROFESSIONAL EXPERIENCE:

Verizon, Irving, TX
Role: Sr. Data Engineer **May 2023 - Present**
Responsibilities:

- ✓ Involved in all phases of Software Development Life Cycle (SDLC) such as requirements gathering, modelling, analysis, design, development, and testing.
- ✓ Developed and deployed Spark applications using Pyspark and Spark-SQL for data extraction, transformation, and aggregation from multiple le formats for analyzing & transforming the data to uncover insights into the customer usage patterns
- ✓ Developed HIVE UDFs to incorporate external business logic into Hive script and developed join data set scripts using HIVE join operations.
- ✓ Created various Hive external tables, staging tables and joined the tables as per the requirement. Implemented static Partitioning, Dynamic partitioning and Bucketing.
- ✓ Worked with the Spark for optimization of the existing algorithms in Hadoop using Spark Context, Spark-SQL, PySpark, Pair RDD's, and Spark YARN.
- ✓ Involved in migrating the client Datawarehouse architecture from on-premises into MS Azure cloud.
- ✓ Used Spark for interactive queries, processing of streaming data and integration with popular NoSQL database for huge volume of data.
- ✓ Worked with NoSQL databases like HBase in creating HBase tables to load large sets of semi-structured data coming from various sources.
- ✓ Created pipelines in ADF using linked services to extract, transform and load data from multiple sources like Azure SQL, Blob storage and Azure SQL Data warehouse.
- ✓ Implemented Cluster for NoSQL tool HBase as a part of POC to address HBase limitations.
- ✓ Used Spark Data Frames Operations to perform required Validations in the data and to perform analytics on the Hive data.
- ✓ Developed Apache Spark applications by using Spark for data processing from various streaming sources.
- ✓ Design and implement end-to-end data solutions (storage, integration, processing, and visualization) in Azure.
- ✓ Worked on architecture and components of Spark, and efficient in working with Spark Core, Spark SQL.
- ✓ Used Kafka and Kafka brokers to initiate spark context and processing livestreaming.
- ✓ Developed custom Kafka producer and consumer for different publishing and subscribing to Kafka topics.
- ✓ Migrated Map reduce jobs to Spark jobs to achieve better performance.
- ✓ Developed Python, Shell/ Perl Scripts and Power shell for automation purpose and Component unit testing using Azure Emulator.
- ✓ Working on designing the MapReduce and Yarn flow and writing MapReduce scripts, performance tuning and debugging.
- ✓ Used HUE for running Hive queries. Created partitions according to day using Hive to improve performance.
- ✓ Implement ad-hoc analysis solutions using Azure Data Lake Analytics/ Store, and HDInsight.
- ✓ Developed Oozie workflow engine to run multiple Hive, Pig, Sqoop and Spark jobs.

- ✓ Implement ETL and data movement solutions using Azure Data Factory (ADF), and SSIS.
- ✓ Worked on auto scaling the instances to design cost effective, fault tolerant and highly reliable systems.
- ✓ ? Develop ETL/ELT pipelines using DBT for data transformations in BigQuery & Snowflake.
- ✓ ? Process large datasets using Databricks (PySpark, Spark SQL) and integrate with Google Cloud Storage (GCS).
- ✓ ? Implement data governance, lineage, and security controls using Databricks Unity Catalog and GCP's IAM & DLP.
- ✓ ? Optimize Snowflake queries, warehouse sizing, and cost management strategies in GCP.
- ✓ ? Automate CI/CD pipelines for DBT and Snowflake using Terraform, Cloud Build, or GitHub Actions.
- ✓ ? Integrate Cloud Data Fusion & Databricks for seamless data movement and transformations.
- ✓ ? Work with structured & semi-structured data (Parquet, JSON, Avro) using BigQuery & Snowflake.

Environment: MS Azure, Azure SQL, Blob Storage, Azure Data Factory (ADF), HDInsight, Python, OLAP, Hadoop (HDFS, MapReduce), YARN, Spark, Spark Context, Spark-SQL, PySpark, Pair RDD's, Spark DataFrames, Spark YARN, Hive, Pig, HBase, Oozie, Hue, Sqoop, Flume, Oracle, NIFI, Kafka ,BigQuery, Snowflake, GCP, Databricks.

Wells Fargo, Charlotte, NC
Role: Data Engineer **Jan 2021 – April 2023**
Responsibilities:

- ✓ Designed and implemented configurable data delivery pipeline for scheduled updates to customer facing data stores built with Python
- ✓ Worked on Ingesting data by going through cleansing and transformations and leveraging AWS Lambda, AWS Glue and Step Functions.
- ✓ Developed HIVE UDFs to incorporate external business logic into Hive script and Developed join data set scripts using HIVE join operations.
- ✓ Exported the analysed data to the relational databases using Sqoop for visualization and to generate reports for the BI team Using Tableau.
- ✓ Implemented Composite server for the data virtualization needs and created multiples views for restricted data access using a REST API.
- ✓ Implemented the machine learning algorithms using python to predict the quantity a user might want to order for a specific item so we can automatically suggest using kinesis firehose and S3 data lake.
- ✓ Allotted permissions, policies and roles to users and groups using AWS Identity and Access Management (IAM).
- ✓ Created various hive external tables, staging tables and joined the tables as per the requirement. Implemented static Partitioning, Dynamic partitioning and Bucketing.
- ✓ Developed custom Kafka producer and consumer for different publishing and subscribing to Kafka topics.
- ✓ Migrated Map reduce jobs to Spark jobs to achieve better performance.
- ✓ Working on designing the MapReduce and Yarn flow and writing MapReduce scripts, performance tuning and debugging.
- ✓ Used Spark-Streaming APIs to perform necessary transformations and actions on the data got from Kafka.
- ✓ Implemented Kafka Connect connectors to integrate Kafka with various external systems, enabling seamless data ingestion and delivery.
- ✓ Developed workflow in Oozie to automate the tasks of loading the data into Nifi and pre-processing with Pig.
- ✓ Worked on Apache NIFI to decompress and move JSON files from local to HDFS.
- ✓ Like Access, Excel, CSV, Oracle, flat files using connectors, tasks and transformations provided by AWS Data Pipeline.
- ✓ Worked with Json format files by using XML, Hierarchical Data stage stages.
- ✓ Extensively used Parallel Stages like Row Generator, Column Generator, Head, and Peek for development and de-bugging purposes.
- ✓ Mentor and guide analyst on building purposeful analytics tables in dbt for cleaner schemas.
- ✓ Developed a python script to transfer data from on-prem with AWS. Worked on the tuning of SQL Queries to bring down run time by working on Indexes and Execution Plan.
- ✓ Reduced analytical query response times and improved query speed by implementing query optimization techniques in Amazon Redshift.
- ✓ Created and executed comprehensive recovery and backup strategies for Amazon Redshift, protecting data and reducing the likelihood of loss.
- ✓ Strong understanding of AWS components such as EC2 and S3
- ✓ Used the Data Stage Director and its run-time engine to schedule running the solution, testing and debugging its components, and monitoring the resulting executable versions on ad hoc or scheduled basis.
- ✓ Stored data in AWS S3 like HDFS and performed EMR programs on data stored.
- ✓ Used the AWS-CLI to suspend an AWS Lambda function. Used AWS CLI to automate backups of ephemeral data-stores to S3 buckets, EBS.
- ✓ If we don't have data on our HDFS cluster, I will be scooping the data from netezza onto out HDFS cluster.
- ✓ Transferred the data using Informatica tool from AWS S3 to AWS Redshift.

- ✓ Worked on Hive UDF's and due to some security privileges I have to ended up the task in middle itself.
- ✓ Implemented a Continuous Delivery pipeline with Docker, and Git Hub and AWS
- ✓ Wrote Flume configuration files for importing streaming log data into HBase with Flume.
- ✓ Experience in setting up the whole app stack, setup, and debug log stash to send Apache logs to AWS Elastic search.
- ✓ Developed Spark code using Scala and Spark-SQL/Streaming for faster testing and processing of data.
- ✓ Implemented AWS provides a variety of computing and networking services to meet the needs of applications
- ✓ Writing HiveQL as per the requirements and Processing data in Spark engine and store in Hive tables.
- ✓ Importing existing datasets from Oracle to Hadoop system using SQOOP.
- ✓ Brought data from various sources in to Hadoop and Cassandra using Kafka.
- ✓ Experienced in using Tidal enterprise scheduler and Oozie Operational Services for coordinating the cluster and scheduling workflows.
- ✓ Model, lift and shift custom SQL and transpose LookML into dbt for materializing incremental views.
- ✓ Applied spark streaming for real time data transforming.

Environment: Hadoop (HDFS, MapReduce), DBT, Data Stage, Scala, Spark, DB2, Snowflake, Impala, Hive, MangoDB, Pig, Devops, HBase, Oozie, Hue, Sqoop, Flume, Oracle, AWS Services (Lambda, EMR, Auto scaling), Mysql, Python, Scala, Spark, Hive, Spark-Sql.

IBX, Pittsburgh, PA
Role: Data Engineer **October 2019 – December 2020**
Responsibilities:

- ✓ Administered, maintained, provisioned, patched, and maintained Cloudera Hadoop clusters on Linux.
- ✓ Worked on analyzing Hadoop stack and different big data analytic tools including Pig and Hive, HBase database, and Sqoop.
- ✓ Utilized Python matplotlib and SciKit-Learn modules to generate the basic prototype visualization restored using visualization tools such as Tableau.
- ✓ Written multiple MapReduce programs to extract data for extraction, transformation, and aggregation from more than 20 sources having multiple file-formats including XML, JSON, CSV & other compressed file formats.
- ✓ Worked in AWS environment for development and deployment of Custom Hadoop Applications.
- ✓ Created Oozie workflows for Hadoop-based jobs including Sqoop, Hive, and Pig.
- ✓ Involved in file movements between HDFS and AWS S3.
- ✓ Created Hive External tables and loaded the data into tables and query data using HQL.
- ✓ Performed data validation on the data ingested using MapReduce by building a custom model to filter all the invalid data and cleanse the data.
- ✓ Handled the importing of data from various data sources, performed transformations using Hive, and Map-Reduce, loaded data into HDFS, and extracted data from MySQL into HDFS using Sqoop.
- ✓ Transferred the data using the Informatica tool from AWS S3 to AWS Redshift.
- ✓ Wrote HiveQL queries by configuring a number of reducers and mappers in the query needed for the output.
- ✓ Transferred data between Pig Scripts and Hive using HCatalog, transferred relational database using Sqoop.
- ✓ Involved in Extract Transfer and Load (ETL) process using SSIS and generated reports using SSRS.
- ✓ Analyzed the data by performing Hive queries (HiveQL) and running Pig Scripts (Pig Latin).
- ✓ Cluster coordination services through Zookeeper. Installed and configured Hive and written Hive UDFs.

Environment: Hadoop, Pig, Hive, HBase, Sqoop, Python, Matplotlib, Scikit Learn, Tableau, HDFS, Map Reduce, AWS, S3, RedShift, MySQL, OLAP, Oozie, Sqoop, HQL, SSIS, SSRS, MS SQL Server, HiveQL.

CVS, Dallas, TX
Data Engineer **June 2017 – September 2019**
Responsibilities:

- ✓ Worked with BI team in gathering the report requirements and Sqoop to export data into HDFS and Hive.
- ✓ Involved in the below phases of Analytics using R, Python, and Jupiter Notebook.
- ✓ Data collection and treatment: Analyzed existing internal data and external data, worked on entry errors, classification errors and defined criteria for missing values.
- ✓ Data Mining: Used cluster analysis for identifying customer segments, Decision trees used for profitable and non-profitable customers, Market Basket Analysis used for customer purchasing behavior and part/ product association.
- ✓ Developed multiple Map Reduce jobs in Java for data cleaning and pre-processing.
- ✓ Assisted with data capacity planning and node forecasting.
- ✓ Installed, Configured and managed Flume Infrastructure.
- ✓ Administrator for Pig, Hive and HBase installing updates patches and upgrades.
- ✓ Worked closely with the claims processing team to obtain patterns in filing of fraudulent claims.

- ✓ Developed Map Reduce programs to extract and transform the data sets and results exported back to RDBMS using Sqoop.
- ✓ Patterns observed in fraudulent claims using text mining in R and Hive.
- ✓ Exported the data required information to RDBMS using Sqoop to make the data available for the claims processing team to assist in processing a claim based on the data.
- ✓ Developed Map Reduce programs to parse the raw data, populate staging tables and store the refined data in partitioned tables in the EDW.
- ✓ Created tables in Hive and loaded the structured (resulted from Map Reduce jobs) data
- ✓ Using Hive QL developed many queries and extracted the required information.
- ✓ Created Hive queries that helped market analysts spot emerging trends by comparing fresh data with EDW reference tables and historical metrics.
- ✓ Responsible for importing the data (mostly log files) from various sources into HDFS using Flume.
- ✓ Enabled speedy reviews and first mover advantages by using Oozie to automate data loading into the Hadoop Distributed File System and PIG to pre-process the data.
- ✓ Managed and reviewed Hadoop log files.
- ✓ Tested raw data and executed performance scripts.

Environment: HDFS, PIG, HIVE, Map Reduce, Linux, HBase, Flume, Sqoop, R, VMware, Eclipse, Cloudera, and Python.

TCS, Hyderabad, India
Hadoop Developer **July 2015 – February 2017**
Responsibilities:

- ✓ Involved in the evaluation of functional and non-functional requirements.
- ✓ Installed and configured Hadoop MapReduce HDFS Developed multiple MapReduce jobs in java for data cleaning and pre-processing.
- ✓ Installed and configured Pig and also written Pig Latin scripts.
- ✓ Wrote MapReduce job using Pig Latin.
- ✓ Involved in managing and reviewing Hadoop log files.
- ✓ Imported data using Sqoop to load data from MySQL to HDFS on regular basis.
- ✓ Developing Scripts and Batch Job to schedule various Hadoop Program.
- ✓ Written Hive queries for data analysis to meet the business requirements and created hive tables and worked on them using Hive QL.
- ✓ Importing and exporting data into HDFS and Hive using Sqoop.
- ✓ Experienced in defining job flows.
- ✓ Got good experience with NOSQL database SOLR HBase.
- ✓ Involved with putting data into Hive tables and constructing hive queries that will execute internally in a map reduce fashion.
- ✓ Created a custom Filesystems plug-in for Hadoop that allows it to access files on the Data Platform.
- ✓ This plugin enables Hadoop MapReduce applications HBase Pig and Hive to function normally and directly access files.
- ✓ Designed and implemented MapReduce-based large-scale parallel relation-learning system.
- ✓ Extracted feeds form social media sites such as Facebook Twitter using Python scripts.
- ✓ For internal purposes, I have set up benchmarked Hadoop clusters.

Environment: Hadoop, MapReduce, HDFS Hive, Java Hadoop distribution of Horton Works, Cloudera, Pig, HBase, Linux, XML, MySQL, MySQL Workbench Java 6, Eclipse, Oracle 10g PL/SQL, SQL PLUS Sub Version Cassandra.